

SILANG WANG

HUMAN COMPUTER INTERACTION RESEARCHER

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SUMMARY OF QUALIFICATIONS

- HCI researcher and developer with 4+ years of experience designing, building, and evaluating AI, XR, and wearable systems. Adept at rapidly learning new domains and collaborating across interdisciplinary and cross-cultural teams to deliver multiple peer-reviewed HCI publications and award-winning prototypes
- Programming: C# (moderate), C/C++ (moderate), Python (moderate), SQL (beginner), Excel VBA (beginner), Java (beginner)
- Languages: English (Bilingual), Mandarin (Native), Singlish (Dialect)

EDUCATION

University of Washington (UW)

Sep 2025 – Jun 2027

Master of Science - Human Centered Design & Engineering (HCDE)

- GPA - 3.9 / 4
- Relevant coursework: Theoretical Foundations of HCDE, User-Centered Design, Usability Studies, Qualitative Research Methods, Physical Computing and Prototyping, Design and Prototyping with Artificial Intelligence, Directed Research Group (VR games design for neurodivergent users)

National University of Singapore (NUS)

Aug 2021 – Jun 2025

Bachelor of Engineering - Computer Engineering (CEG)

- GPA - 4.55 / 5 (Highest Distinction Honor)
- Relevant course work – CEG Capstone Project, Machine Learning, Introduction to Artificial Intelligence, Software Engineering & Object-Oriented Programming, Introduction to Psychology, Social Psychology of New Media, Design Thinking, Interaction Design

RELEVANT EXPERIENCES

HCI Research Intern, NUS-HCI Lab, Singapore

Jun 2022 - Jun 2025

- Led an international hybrid team of five undergraduate and PhD researchers in publishing a first-author peer-reviewed publication on a multimodal AI-powered, in-situ information retrieval system on AR glasses via 13 mixed method pilot studies and three iteratively designed VLM/LLM- and AR-based prototypes
- Co-authored two more peer-reviewed publications on mobile messaging and interactive multimodal information retrieval on AR glasses via literature review, user studies, data analysis, and academic writing
- Mentored 40 Chinese participants in their interaction design processes during the Huawei Smartglasses Hackathon in Zhejiang University, China, helping three teams attain a prize in the end

Co-founder, YI KUAN SPORTS PTE. LTD., Singapore

Jun 2025 - Aug 2025

- Designed and launched two mixed-reality sports class pilots at Singapore's National Stadium, gathering early user feedback from 10 participants to refine the product concept and sports curriculum
- Drove early go-to-market outreach, initiating partnership discussions with three local schools at secondary and tertiary levels and Singapore's Minister of Education to explore adoption of mixed-reality sports programs in formal PE settings

RELEVANT PROJECTS

Luce, HCDE 539 Physical Computing Project, UW HCDE, Seattle

Mar 2026 – Jun 2026

- Designed and built a connected book lamp system inspired by Tangible Bits, Bao Bao Issey Miyake and Lumio Book Lamp, transforming the act of opening one book into synchronous illumination across all connected books for gracious telepresence
- Self-taught laser cutting and 3D printing from zero prior physical fabrication experience, earning nearly full score on all deliverables and strong commendation from instructor and TA
- Iterated through two switch designs, four power supply configurations, and multiple exterior, hinge, and fold fabrication approaches to produce the only fully standalone and portable project in the HCDE 539 cohorts

Dream.ee, *Runner-up of Founder's Lab Track, MIT Reality Hack, MIT*

Jan 2026

- Created an EEG to VR pipeline for updating 3D AI-generated environments in real-time based on the user's level of concentration, using Unity, BrainBit 2 wearable EEG BCI and Marbles from World Lab
- Demoed the prototype to 20+ judges and sponsors at the hackathon across academics and industry, winning the runner-up in the Founder's Lab (best start-up idea) track

Anu, *Winner of Meta Track and VR Haptics Track, Immerse the Bay Hackathon, Stanford*

Nov 2025

- Developed an interactive digital legacy prototype with VR+AI+haptics on Meta Quest 3 in 36 hours, where users can experience tangible videos and engage in mediated and ethical Gen-AI based communication with the deceased in VR, using Unity, OpenAI and Afference haptics Ring
- Demoed the prototype to 20+ judges and sponsors at the hackathon across academics and industry, winning the honorable mention in the Meta track and 2nd place in the Afference Ring (VR haptics) track

EEG-VR Sophrologist, *Student Research Project, NUS Computing Dean's Office, Singapore* **Jun 2025 - Aug 2025**

- Developed an EEG-VR neurofeedback meditation application on Meta Quest 3, where users can practice focused attention mediation (FAM) in VR, and evaluate its effectiveness in real time
- Created two virtual environments with different multimodal stimuli for practising FAM in VR using Unity and C#
- Developed an end-to-end MQTT communication pipeline between the EEG headset and Meta Quest 3 VR headset to enable real time neurofeedback during meditation with OpenBCI GUI, Python and Unity

CG4002 CEG Capstone Project, *Only AR Glasses Project in Cohort, NUS, Singapore*

Jan 2025 - May 2025

- Developed an AR+AI visualizer on Microsoft HoloLens 2 for a multiplayer AR laser-tag game, with a LLM-based game tutor using Unity and C#
- Integrated the visualizer with inertial measurement unit (IMU) sensors and Arduino Beetle worn by the players for game action input, via MQTT communication protocol over Wi-fi and an intermediate MQTT broker laptop
- Collaborated with four NUS computer engineering undergraduates working on other components of the system to produce the only laser tag project with an AR head-mounted display visualizer in our cohort, featured in Channel News Asia as part of a documentary on innovation at NUS